

SQL Healthcare Project

Covers hospital admission and discharge data from Hero DMC Heart Institute

Downloaded from <https://www.kaggle.com/datasets/ashishsahani/hospital-admissions-data>

```
-- Point the script to the correct database
```

```
USE AutoCareDB
```

```
GO
```

```
;
```

```
-- Create a view from CleanData
```

```
CREATE OR ALTER VIEW vw_AdmissionData AS
```

```
-- Creating a CTE(a temp table) called 'CleanData' to extract non-duplicate data
```

```
WITH CleanData AS (
```

```
    -- Selecting all the data to identify duplicate using 'Dup_NO' column generated by  
    window function
```

```
    SELECT *,
```

```
    ROW_NUMBER() OVER (PARTITION BY MRD_No, D_O_A, D_O_D ORDER BY MRD_No) AS
```

```
Dup_NO
```

```
    FROM [HDHI Admission data]
```

```
    -- ORDER BY MRD_No
```

```
)
```

```
-- Select non-duplicate data
```

```
SELECT *
```

```
FROM CleanData
```

```
WHERE Dup_NO = 1 AND MRD_No IS NOT NULL
```

```
-- ORDER BY MRD_No
```

```
;
```

```
GO
```

```
-- 1.) Total Discharges
```

```
SELECT COUNT(*) AS Total_Discharges
```

```
FROM vw_AdmissionData
```

```
WHERE OUTCOME = 'DISCHARGE'
```

```
;
```

```
-- 2.) Average Daily Discharge Rate
```

```
-- Total discharges divided by total length of stay
```

```
SELECT
```

```
(SELECT COUNT(*) AS Total_Discharges
```

```
FROM vw_AdmissionData
```

```
WHERE OUTCOME = 'DISCHARGE')/
```

```
(SELECT SUM(DURATION_OF_STAY) AS Total_Length_of_Stay
FROM vw_AdmissionData)
;
```

-- Casting this

```
SELECT
    CAST(
        CAST((SELECT COUNT(*) AS Total_Discharges
FROM vw_AdmissionData
WHERE OUTCOME = 'DISCHARGE') AS FLOAT)/
        CAST((SELECT SUM(DURATION_OF_STAY) AS Total_Length_of_Stay
FROM vw_AdmissionData) AS FLOAT)
        AS DECIMAL(10, 2)) * 100 AS Avg_DailyDischargeRate
;
```

-- Avoiding Subquery

```
SELECT
    ROUND(SUM(CASE WHEN OUTCOME = 'DISCHARGE' THEN 1.0 ELSE 0.0 END)/
        SUM(DURATION_OF_STAY), 2) * 100 AS Avg_DailyDischargeRate
FROM vw_AdmissionData
;
```

-- 3.) Average Length of Stay (ALOS)

-- This is the total length of stay divided by Total Discharges.

-- This is also the inverse of Question 2.

```
SELECT ROUND(SUM(DURATION_OF_STAY)/SUM(CASE WHEN OUTCOME = 'DISCHARGE' THEN
1.0 ELSE 0.0 END), 0) AS Avg_length_of_stay
FROM vw_AdmissionData
;
```

-- 4.) Distribution of discharges by Age Group

-- Pediatric = under 16

-- Adult = 16 to 64 inclusive

-- Seneior Citizen = 65 and older

```
SELECT CASE
    WHEN AGE < 16 THEN 'Pediatric'
    WHEN AGE < 65 THEN 'Adult'
    WHEN 65 <= AGE THEN 'Senior Citizen'
    ELSE 'Unknown'
END AS Age_Group, COUNT(*) AS Age_Distribution
FROM vw_AdmissionData
WHERE OUTCOME = 'DISCHARGE'
GROUP BY CASE
    WHEN AGE < 16 THEN 'Pediatric'
```

```

        WHEN AGE < 65 THEN 'Adult'
        WHEN 65 <= AGE THEN 'Senior Citizen'
        ELSE 'Unknown'
    END
ORDER BY 2 DESC
;

-- 5.) Distribution of discharges by Gender
SELECT GENDER, COUNT(*) AS Gender_Distribution
FROM vw_AdmissionData
WHERE OUTCOME = 'DISCHARGE'
GROUP BY GENDER
ORDER BY 2 DESC
;

-- 6.) Distribution of discharges by day of the week
SELECT DATEPART(WEEKDAY, D_O_D) AS Day_of_Week, COUNT(*) AS Day_Distribution
FROM vw_AdmissionData
WHERE OUTCOME = 'DISCHARGE'
GROUP BY DATEPART(WEEKDAY, D_O_D)
-- GROUP BY DATEPART(WEEKDAY, D_O_D)
ORDER BY 2 DESC
;

-- Get Date Name
SELECT FORMAT(D_O_D, 'ddd') AS Day_of_Week, COUNT(*) AS Day_Distribution
FROM vw_AdmissionData
WHERE OUTCOME = 'DISCHARGE' AND D_O_D IS NOT NULL
GROUP BY FORMAT(D_O_D, 'ddd')
ORDER BY 2 DESC
;

```